UCCD2303 DATABASE TECHNOLOGY / UCCD2203 DATABASE SYSTEMS

Introduction to Data Modelling and Design

Part 1

Lecture 1

By

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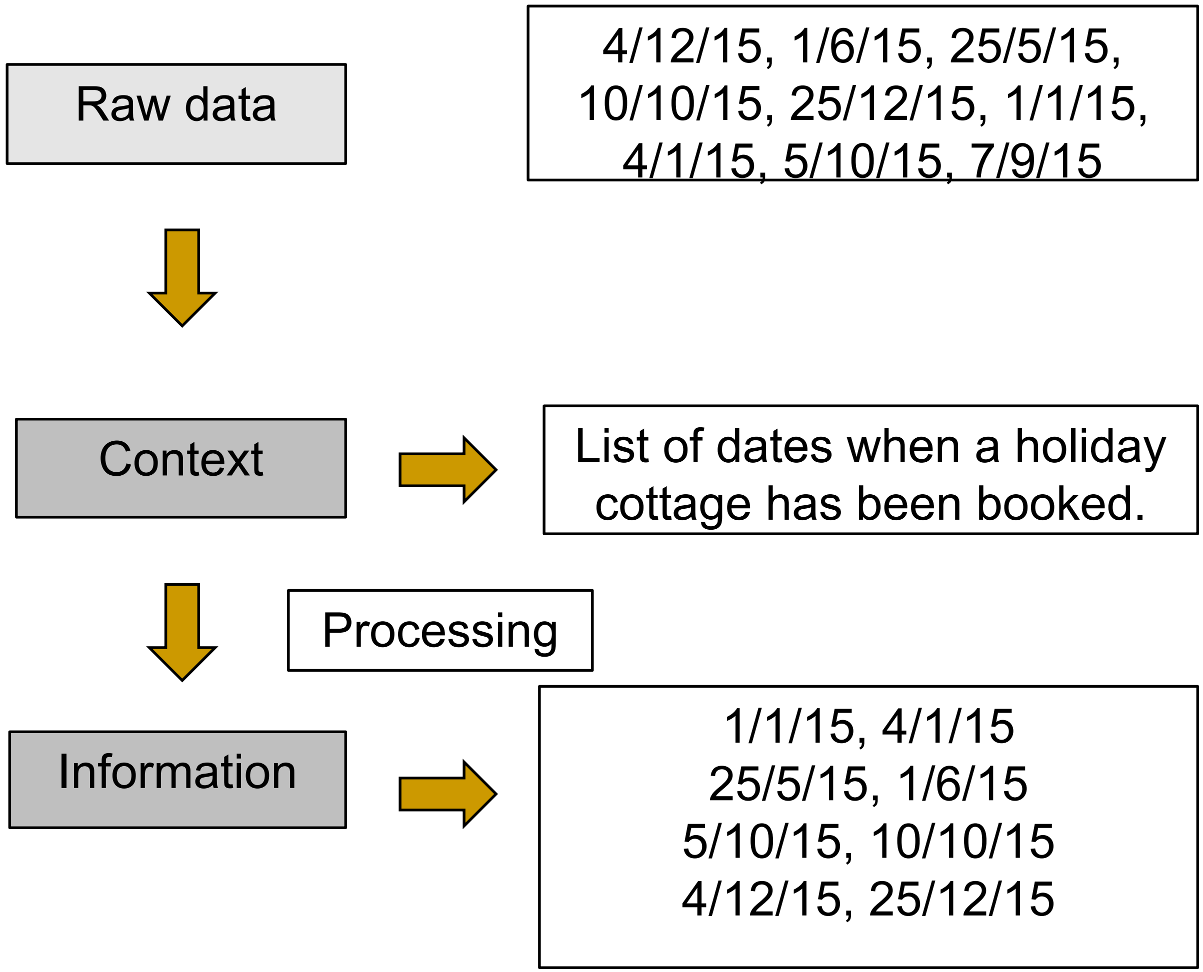
Lecture 1 Topic: Introduction to Data Modelling and Design – Part 1

- Database Concepts
- Database Architecture
- Data Modelling Using the Entity Relationship (ER) Model

DATABASE CONCEPTS

Data Vs Information Vs Knowledge

- **Data :** raw material, from which one can deduce new facts.
- **Information:** processed data (Summarized, Organized, Labeled), meaningful to audience.
- **Knowledge:** Knowing what information is required and knowing what the information means.

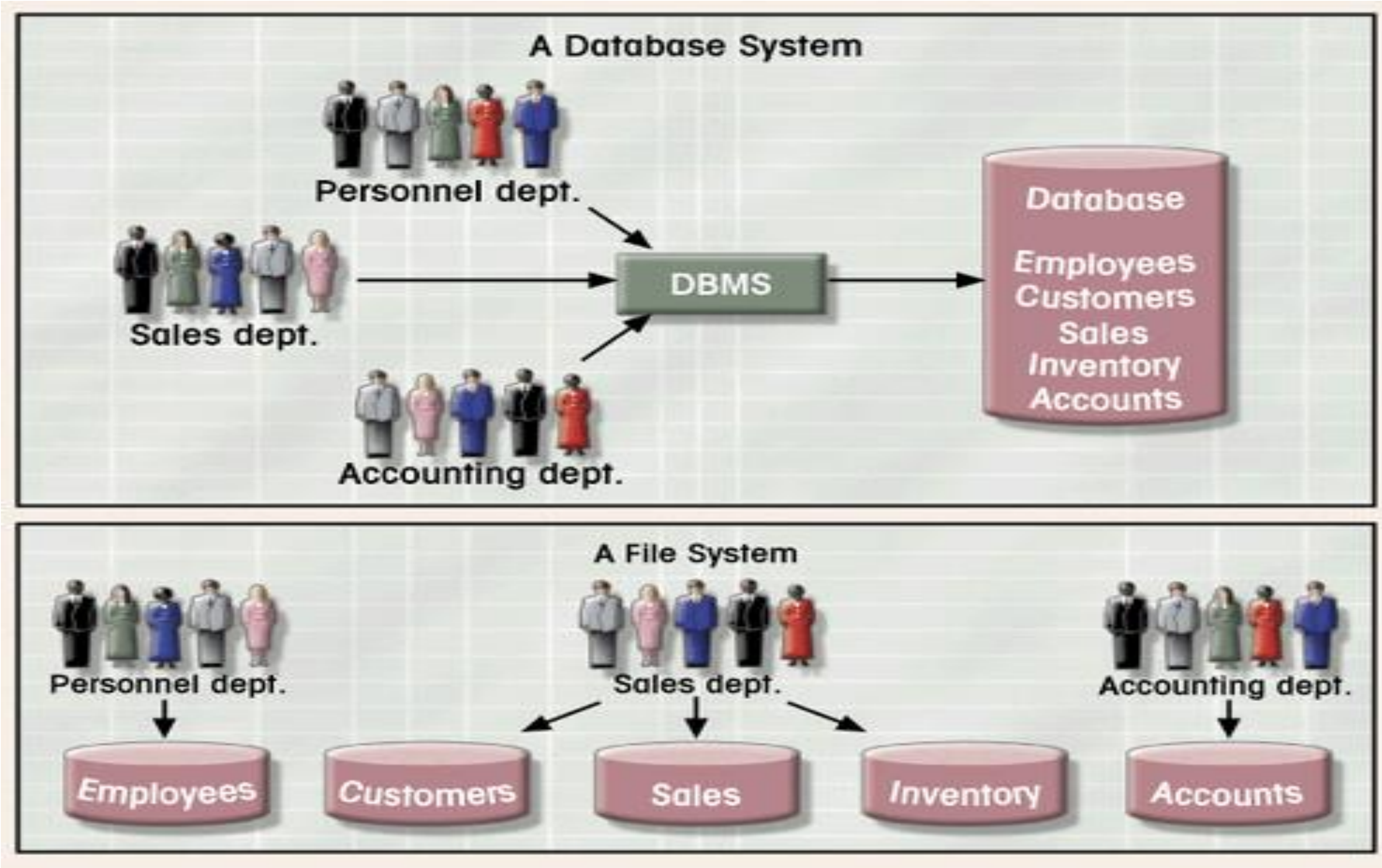


Raw data
4/12/15, 1/6/15, 25/5/15, 10/10/15, 25/12/15, 1/1/15, 4/1/15, 5/10/15, 7/9/15
Context
List of dates when a holiday cottage has been booked.
Processing
Information
1/1/15, 4/1/15
25/5/15, 1/6/15
5/10/15, 10/10/15
4/12/15, 25/12/15

Limitations of File-based Approach

- Separation and isolation of data
- Duplication of data
- Data dependence
- Incompatible file formats
- Fixed Queries/Proliferation (increase) of application programs

Database System vs. File System



Database

- A shared collection of logically related data (raw facts) and a description of this data, designed to meet the information need of an organization.

- Metadata (data about data)

End user data

FIGURE 1.3 CONTENTS OF THE CUSTOMER FILE

C_NAME	C_PHONE	C_ADDRESS	C_ZIP	A_NAME	A_PHONE	TP	AMT	REN
Alfred A. Ramas	615-844-2573	218 Fork Rd., Babs, TN	36123	Leah F. Hahn	615-882-1244	T1	$100.00	05-Apr-2004
Leona K. Dunne	713-894-1238	Box 12A, Fox, KY	25246	Alex B. Alby	713-228-1249	T1	$250.00	16-Jun-2004
Kathy W. Smith	615-894-2285	125 Oak Ln, Babs, TN	36123	Leah F. Hahn	615-882-2144	S2	$150.00	29-Jan-2005
Paul F. Olowski	615-894-2180	217 Lee Ln., Babs, TN	36123	Leah F. Hahn	615-882-1244	S1	$300.00	14-Oct-2004
Myron Orlando	615-222-1672	Box 111, New, TN	36155	Alex B. Alby	713-228-1249	T1	$100.00	28-Dec-2004
Amy B. O'Brian	713-442-3381	387 Troll Dr., Fox, KY	25246	John T. Okon	615-123-5589	T2	$850.00	22-Sep-2004
James G. Brown	615-297-1228	21 Tye Rd., Nash, TN	37118	Leah F. Hahn	615-882-1244	S1	$120.00	25-Mar-2004
George Williams	615-290-2556	155 Maple, Nash, TN	37119	John T. Okon	615-123-5589	S1	$250.00	17-Jul-2004
Anne G. Farriss	713-382-7185	2119 Elm, Crew, KY	25432	Alex B. Alby	713-228-1249	T2	$100.00	03-Dec-2004
Olette K. Smith	615-297-3809	2782 Main, Nash, TN	37118	John T. Okon	615-123-5589	S2	$500.00	14-Mar-2004

C_NAME = Customer name
C_PHONE = Customer phone
C_ADDRESS = Customer address
C_ZIP = Customer ZIP code

A_NAME = Agent name
A_PHONE = Agent phone
TP = Insurance type
AMT = Insurance policy amount, in thousands of $
REN = Insurance renewal date

Table 1-1 Example Metadata for Class Roster

Data Item		*Value*				
Name	**Type**	**Length**	**Min**	**Max**	**Description**	**Source**
Course	Alphanumeric	30			Course ID and name	Academic Unit
Section	Integer	1	1	9	Section number	Registrar
Semester	Alphanumeric	10			Semester and year	Registrar
Name	Alphanumeric	30			Student name	Student IS
ID	Integer	9			Student ID (SSN)	Student IS
Major	Alphanumeric	4			Student major	Student IS
GPA	Decimal	3	0.0	4.0	Student grade point average	Academic Unit

Descriptions of the properties or characteristics of the data, including data types, field sizes, allowable values, and data context

Database Management Systems (DBMS)

- A software system that enables users to define, create, maintain and control access to the database.
  - Collection of programs that manages database structure and controls access to data
  - Possible to share data among multiple applications or users
  - Makes data management more efficient and effective

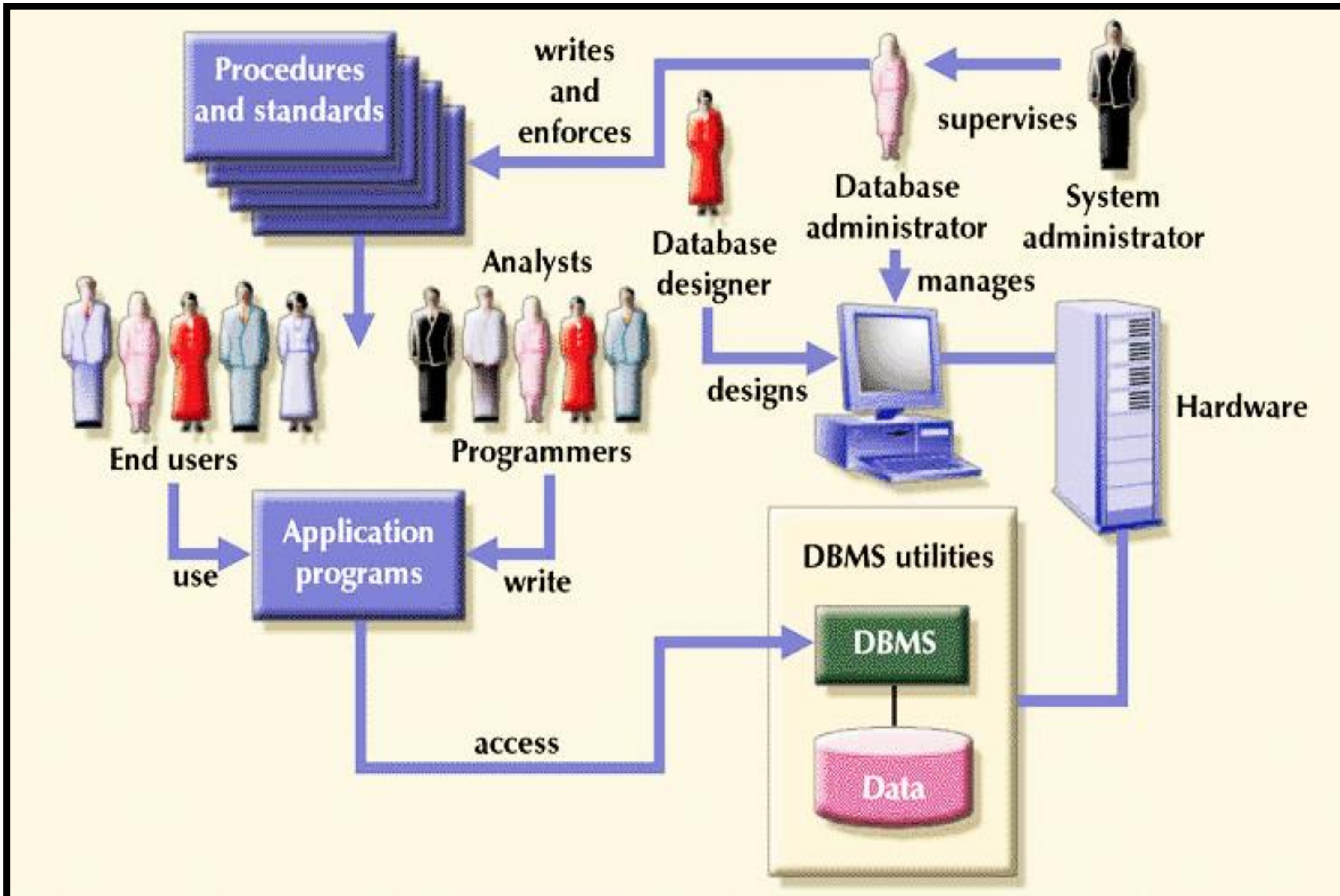
Database Characteristics

- **Persistent**
  - Lasts a long time, outlives program that operate it.
  - Relevance of intended usage: only store potentially relevant data
- **Inter-related**
  - Entity: cluster of data about a topic (course, student, loan)
  - Relationship: connection among entities
- **Shared**
  - *Multiple uses*: hundreds to thousands of data entry screens and reports
  - *Multiple users*: many people simultaneously use a database
- **Reliability**
  - up all the time

Database Characteristics

- **Massive**
  -hundreds of gigabytes for medium size organization
- **Safe**
  - from system/software failure, from malicious users
- **Efficient**
  - retrieval, storing, etc
- **Convenient**
  - Simple commands to get the information

Database System Environment



Roles in the Database Environment

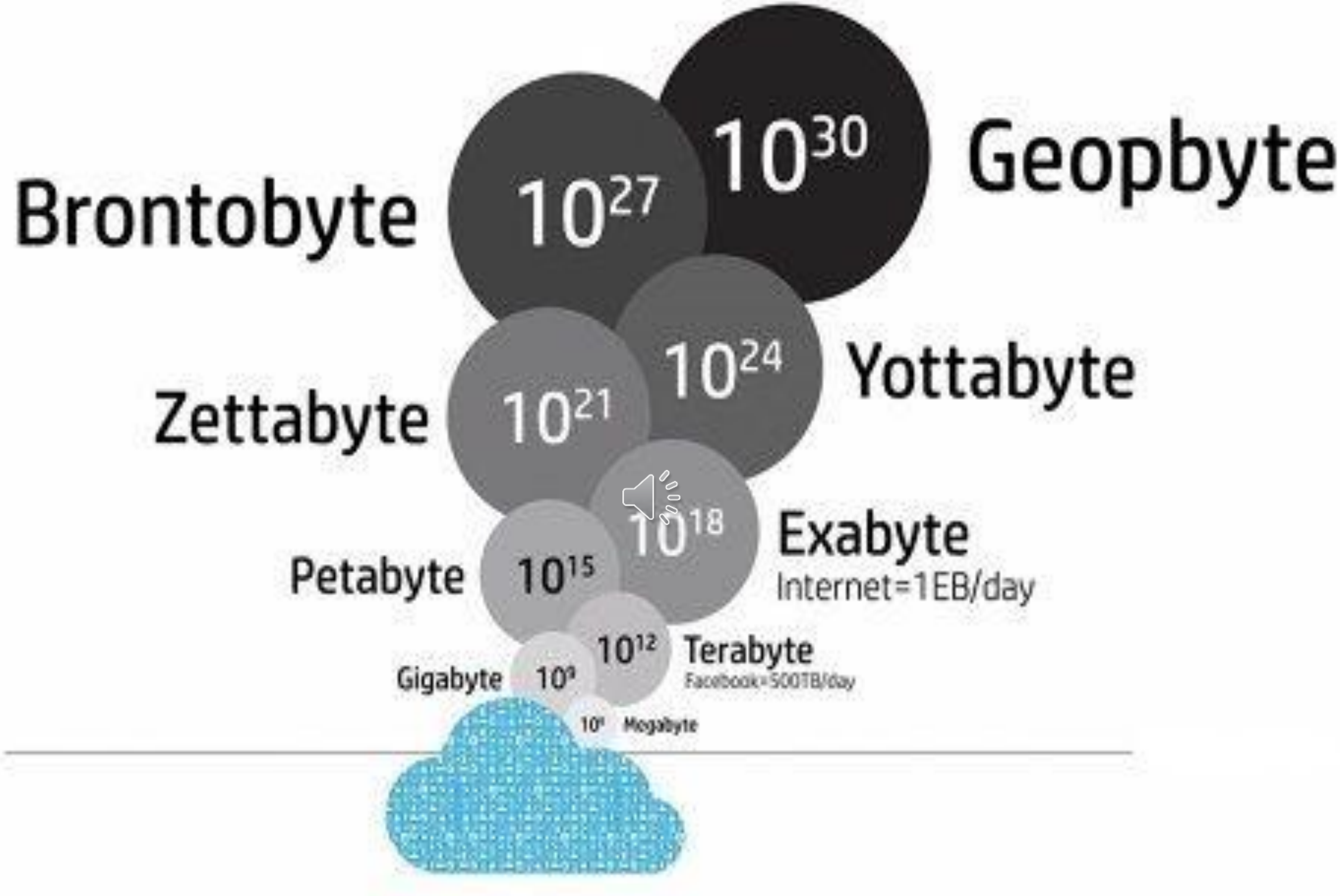
- **Data and Database administrator**
  - Database administrator (DBA): More technical, DBMS specific skills
  - Data administrator: Less technical, Planning role
- **Database Designers**
  - Logical Database Designer: concerned on identifying the data, its relationship and constraints, ways to store in database.
  - Physical Database Designer: decides how the logical DB design is to be physically realized
- **Application Developers:** Works from a specification provided by system analysts.
- **End-Users**
  - Naïve users: are typically unfamiliar with DBMS and
  - Sophisticated users: are familiar with the structure and facilities of DBMS

Advantages of DBMS

- Control of data redundancy
- Data consistency
- More information from the same amount of data.
- Sharing of data
- Improved data integrity (validity & consistency)
- Improved security
- Enforcement of standards
- Economy of scale (cost saving, shared resources)
- Balance of conflicting requirements
- Improved data accessibility and responsiveness
- Increased productivity (reduced programming time)
- Improved maintenance through data independence
- Increased concurrency
- Improved backup and recovery services

Disadvantages of DBMS

- Complexity (all functionalities make its design very complex as well as s/w)
- Size (complexity and many functions add to size, MBs storage and memory space required)
- Cost of DBMSs
- Additional hardware costs
- Cost of conversion (system, training, new staff hiring)
- Performance (may be slower in certain cases)
- Greater impact of a failure(centralized resources, increased vulnerability)



10^{30}
Geopbyte
Brontobyte
10^{27}
10^{24}
Yottabyte
Zettabyte
10^{21}
10^{18}
Exabyte
Internet=1EB/day
Petabyte
10^{15}
10^{12}
Terabyte
Facebook=500TB/day
Gigabyte
10^9
Megabyte

Database Terminology

Database: A repository where an ordered and integrated collection of the enterprise data is stored for computer processing and information sharing.

Database Management System (DBMS): DBMS is a software system that enables users to define, create, maintain and control access to the database.

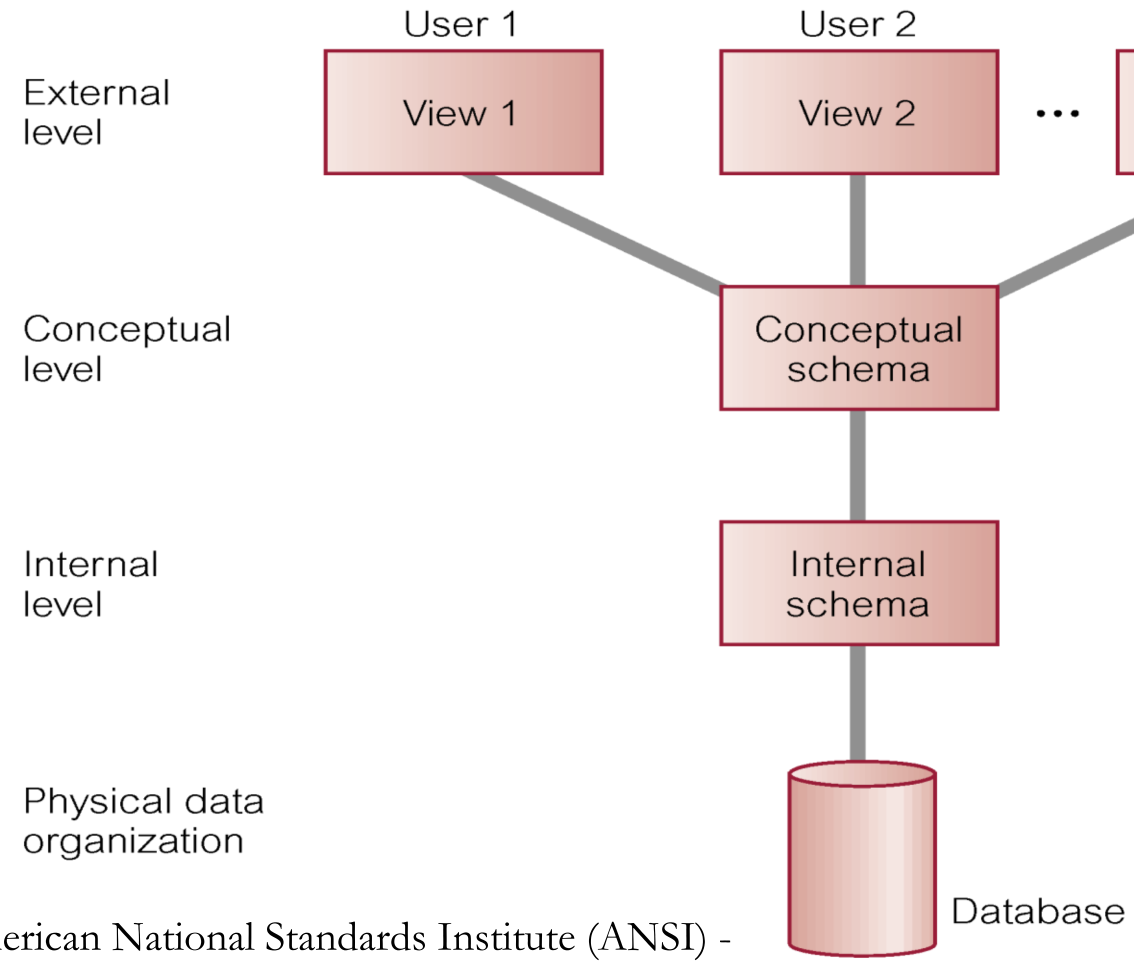
Database System: The combination of a database, a database management system and a data model. It is responsible for the following data manipulation acts; data controlling, data retrieving, data maintenance and data definition.

Database Terminology

Data Warehouse (DW): It is a collection of methods, techniques, and tools used to support knowledge workers-senior managers, directors, and analysts to conduct data analyses that help with performing decision-making processes and improving information resources.

Data Mining (DM): A data-driven approach to analysis and prediction by applying sophisticated techniques and algorithms to discover knowledge.

Three-level ANSI-SPARC Architecture*



* The American National Standards Institute (ANSI) - Standards Planning and Requirements Committee (SPARC)

Refer: Fig 2.1, Page: 87

Objectives of Three-Level Architecture

- It allows independent customized user views
- It hides the physical storage details from users
- The database administrator should be able to change the database storage structures without affecting the users’ views
- The internal structure of the database should be unaffected by changes to the physical aspects of the storage

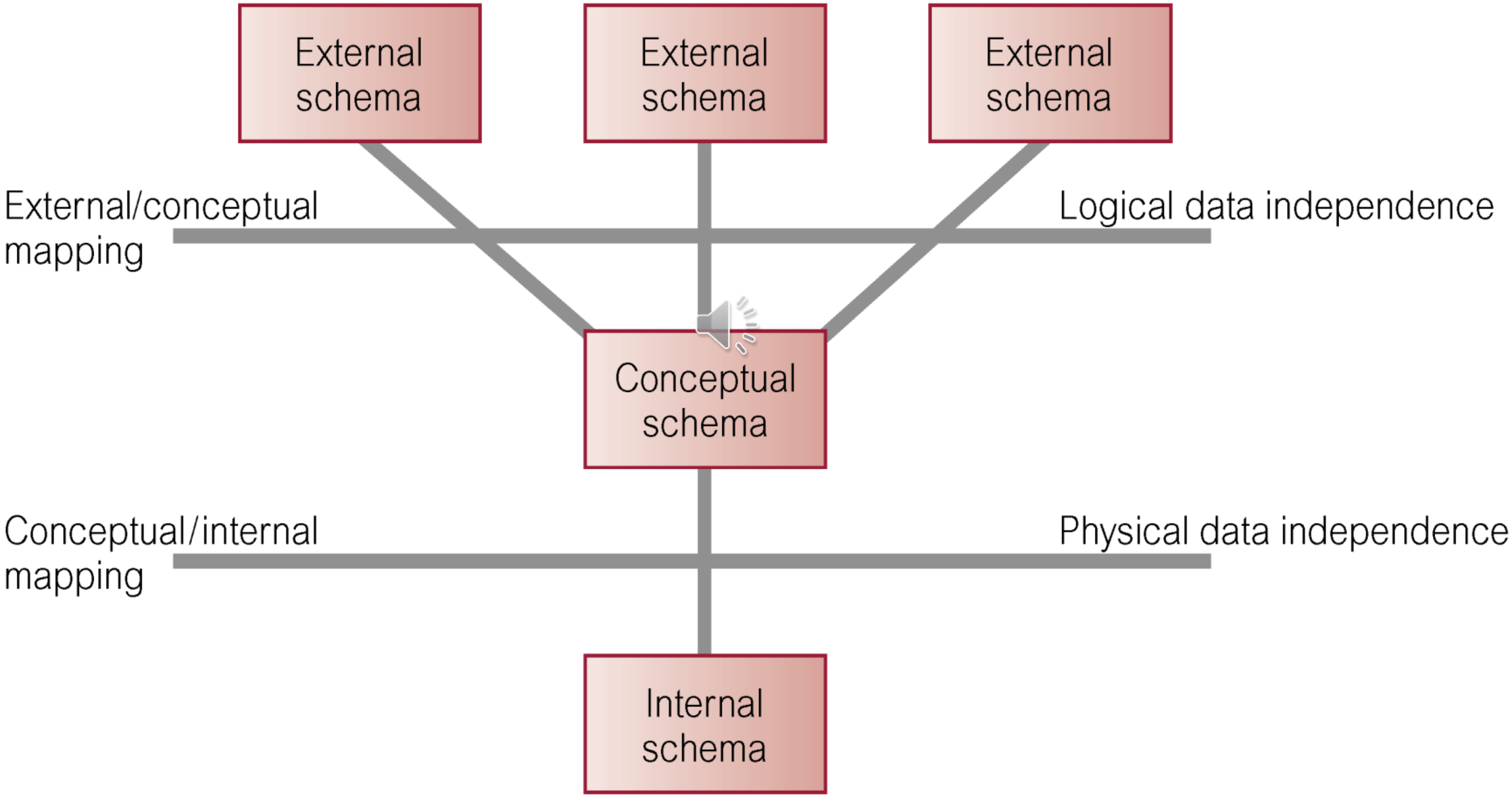
Three-level ANSI-SPARC Architecture

1. **External Level**
   - Users' view of the database.
   - Describes that part of database that is relevant to a particular user.
2. **Conceptual Level**
   - Community view of the database.
   - Describes what data is stored in database and relationships among the data.
3. **Internal Level**
   - Physical representation of the database on the computer.
   - Describes how the data is stored in the database.
   - Depends on specific database software
     - Change in DBMS software requires internal model be changed

Schemas, Mapping and Instances

- **Database Schema** (intension of the database)
  - ✓ The description of the database.
  - ✓ Specified during design process.
  - ✓ Do not change frequently unlike actual data.
- **Database Instance** (extension/state of database)
  - ✓ The data in database at particular point in time.
  - ✓ Same DB schema – many instances.Describes that part of database that is relevant to a particular user.
- **Conceptual/Internal mapping**
  - ✓ to find actual record in physical storage - logical record in conc. schema.
- **External/Conceptual mapping**
  - ✓ to map names in user's view to the relevant part of the conc. schema.

Data Independence and the ANSI-SPARC Three-level Architecture



Database Languages

- **Data Definition Language** (DDL)
  - Allows the DBA or user to describe and name entities, attributes, and relationships required for the application.
  - Used to define a schema or modify an existing one.
  - plus any associated integrity and security constraints.
- **Data Manipulation Language** (DML)
  - Provides basic data manipulation operations on data held in the database e.g. Insert, delete, update records.
  - <u>Procedural DML</u>: allows user to tell system exactly how to manipulate data. Embedded in high level languages.
  - <u>Non-Procedural DML</u>: allows user to state what data is needed rather than how it is to be retrieved.

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Database Languages

- **Fourth Generation Language (4GL)**

  4GL are non-procedural languages, but define parameters for the tools that uses them to generate an application program.

- **<u>Forms Generators:</u>**
  - ✓ an interactive facility for rapidly creating data input and display outputs for screen forms.
- **<u>Report Generators:</u>**
  - ✓ retrieves stored data from database and create reports.
- **<u>Graphics Generators</u>:**
  - ✓ retrieves data from database and display the data as a graph.
- **<u>Application Generators</u>:**
  - ✓ that define, insert, update and retrieve data from the database to build applications.

Data Models

- Data Model

  Integrated collection of concepts for describing data, relationships between data, and constraints on the data in an organization.

  - **Object-based Data Models**
    - Entity-Relationship
    - Semantic
    - Functional
    - Object-Oriented
  - **Record-based Data Models**
    - Relational Data Model
    - Network Data Model
    - Hierarchical Data Model
  - **Physical Data Models**

Data Models and Conceptual Modeling

- **Conceptual Modeling** (Conceptual DB Design)

- ✓ Conceptual modeling is process of developing a model of information use that is independent of implementation details.
- ✓ Result is a conceptual data model.
- ✓ Conceptual schema is the core of a system supporting all user views.
- ✓ Independent of implementation.
- ✓ Should be complete and accurate representation of an organization's data requirements.

DBMS Functions

- Data storage, retrieval, and update
- A user-accessible catalog
- Transaction support
- Concurrency control services
- Recovery services
- Authorization services
- Support for data communication
- Integrity services
- Services to promote data independence
- Utility services – import , monitor, analysis, garbage collection

DATABASE ARCHITECTURE

DATABASE ARCHITECTURE

- Multi-user DBMS Architectures
- Web services and Service-oriented Architectures
- Distributed DBMS
- Data Warehousing

1. Multi-User DBMS Architectures

1.1 Teleprocessing

1.2 File-Server Architecture

1.3 Two-Tier Client-Server Architecture

1.4 Three-Tier Client-Server Architecture

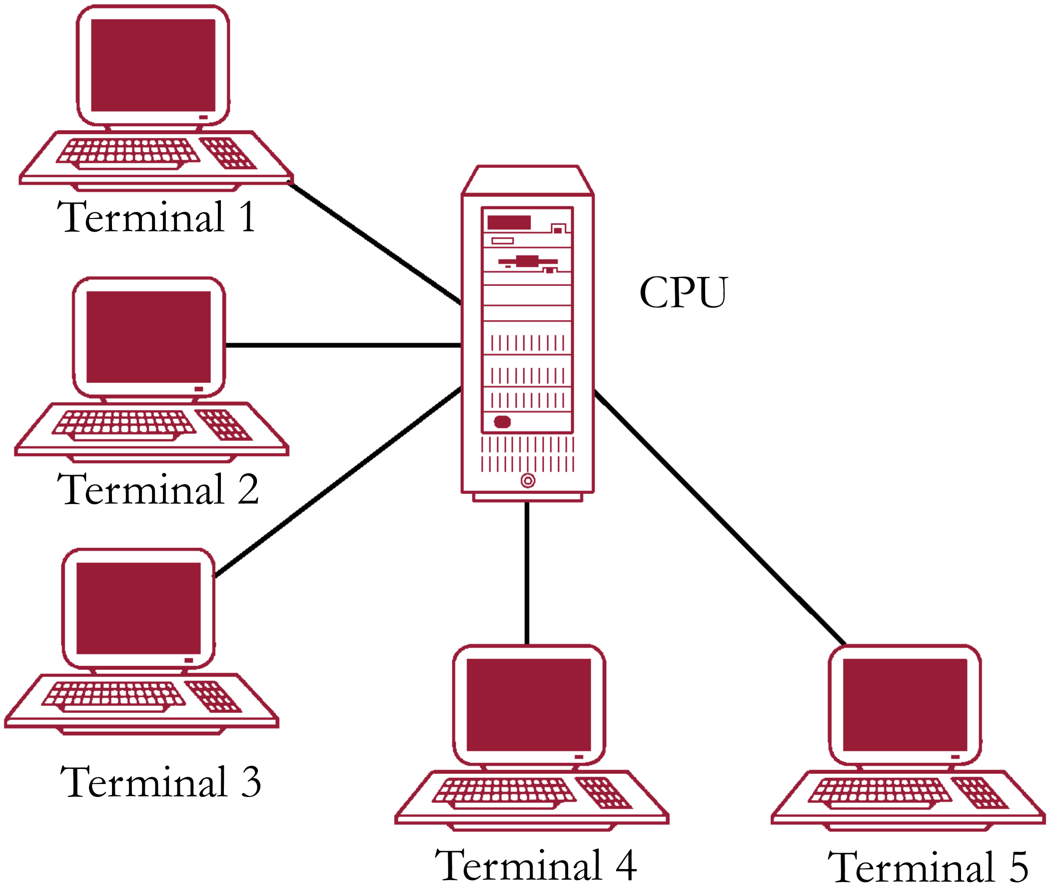
1.5 *N*-Tier Architectures

1.6 Transaction Processing Monitors

1.1 Teleprocessing (1970s)

- Traditional architecture.
- Single mainframe with a number of terminals attached.
- The terminals depended on the mainframe
- The database software resided in the mainframe computer. Almost all computer processing was done on these large mainframe computers.

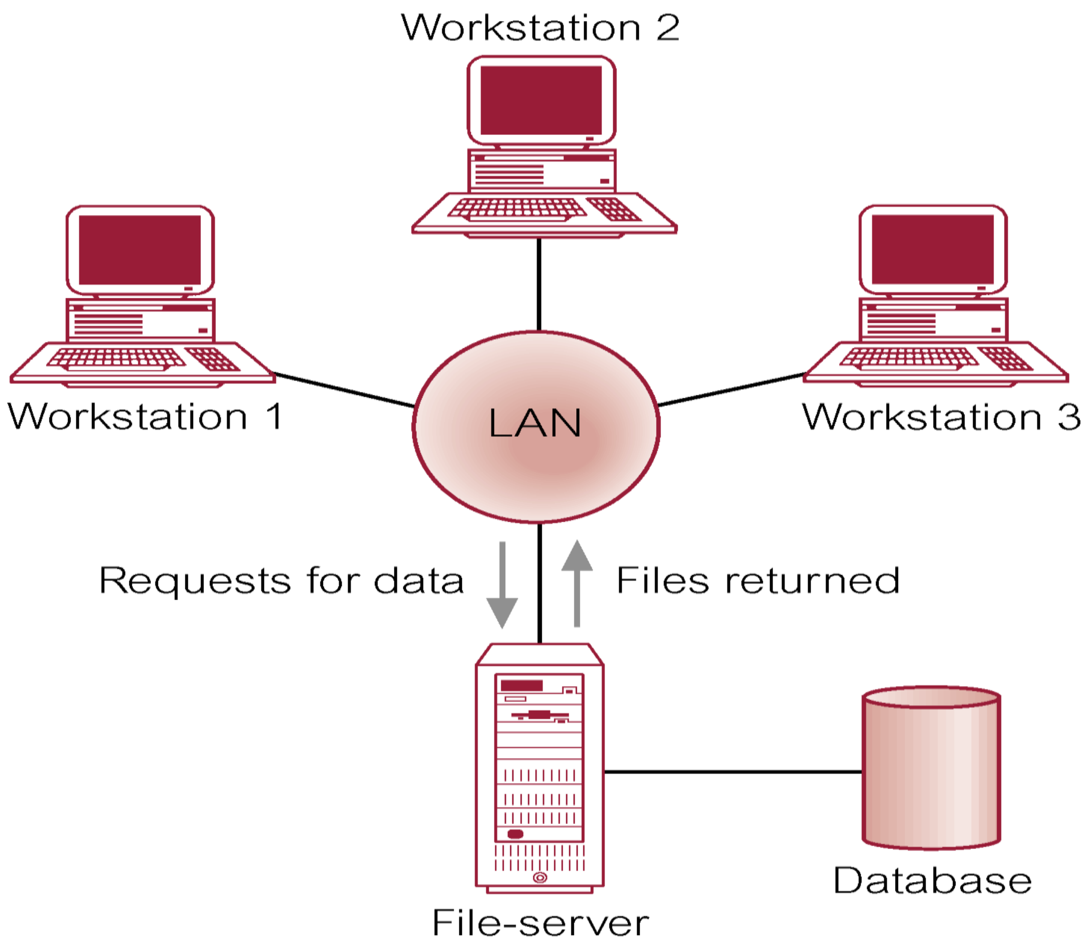
Teleprocessing Topology



1.2 File-Server (1980s)

- File-server is connected to several workstations across a network.
- Database resides on file-server.
- DBMS and applications run on each workstation.
- Disadvantages include:
  - Significant network traffic.
  - Copy of DBMS on each workstation.
  - Concurrency, recovery and integrity control more complex.

File-server Architecture



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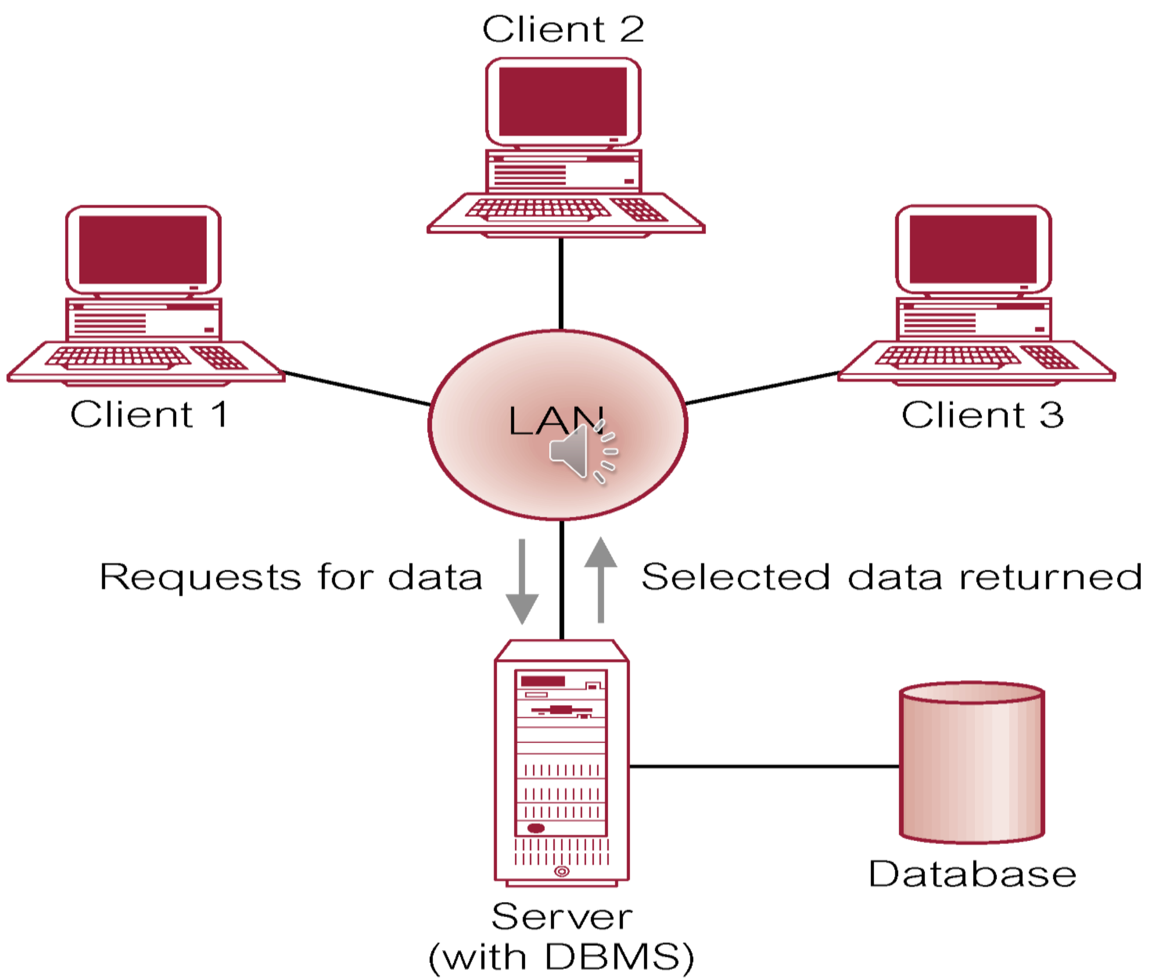
1.3 Two-Tier Client-server Architecture (1990s)

- Server holds the database and the DBMS.
- Client manages user interface and runs applications.
- Uses the internet and fast processing servers to meet the needs.
- Advantages include:
  - wider access to existing databases
  - increased performance
  - possible reduction in hardware/software costs
  - reduction in communication costs
  - increased consistency.

Two-Tier Client-server Architecture



Client-server Architectures

Generally a DBMS is divided into *two* parts:

- **Client** – program that handles the main business and data processing logic and interfaces with the user;
- **Server** – program that manages and control access to the database

~~ two-tier architecture/ client-server architecture

Variation of Client-server Architectures

1.4 Three-Tier architecture

Client server (User interface layer)

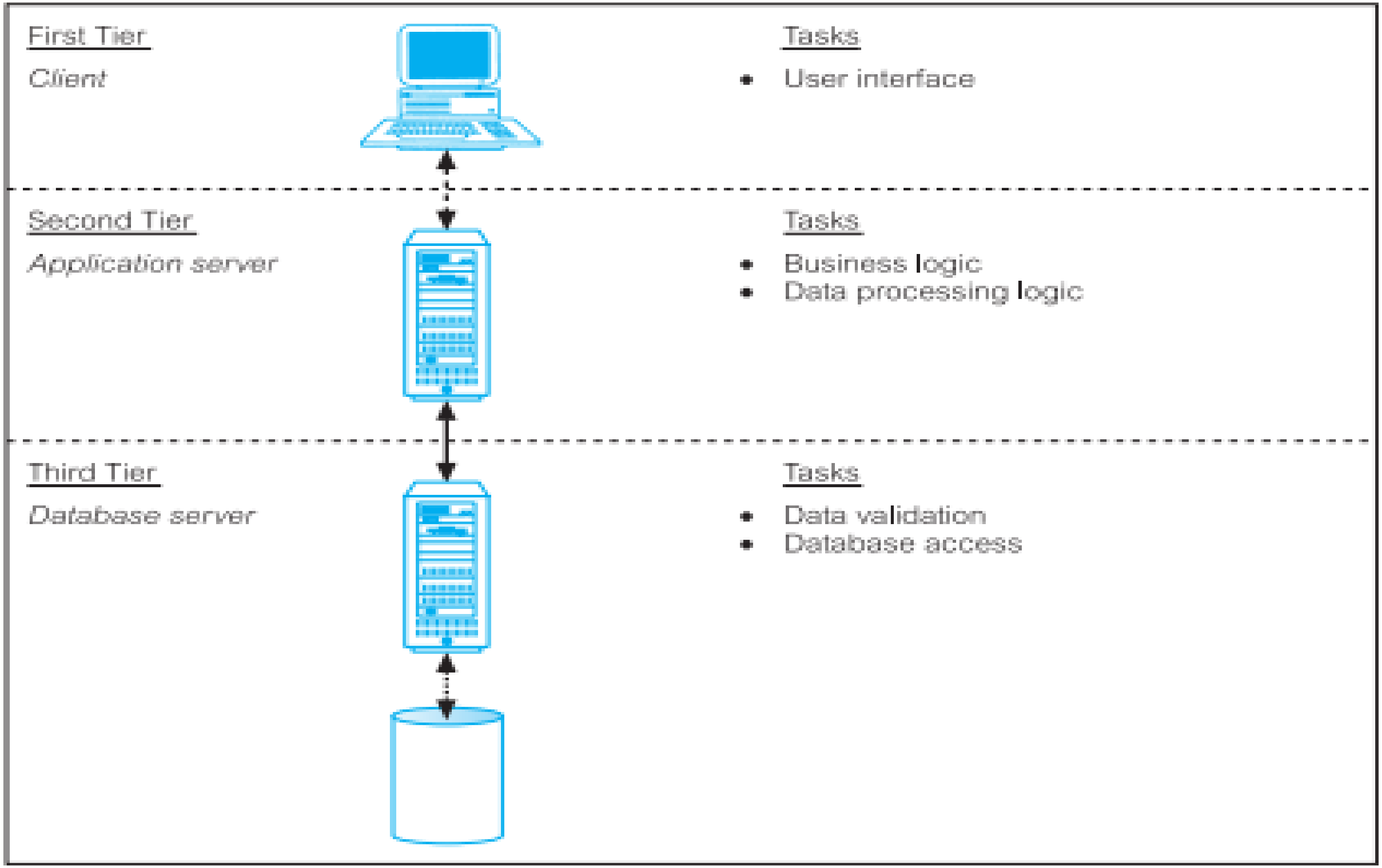
Application server (Business logic and data processing layer)

Database server (DBMS/storage)

1.5 Four-Tier architecture (*N*-Tier)

- Three-Tier architecture can be expanded to *N*-Tier
    1. Client (User interface layer)
    2. Application server (Business logic and data processing layer)
    3. Database server (DBMS/storage)
    4. Web server

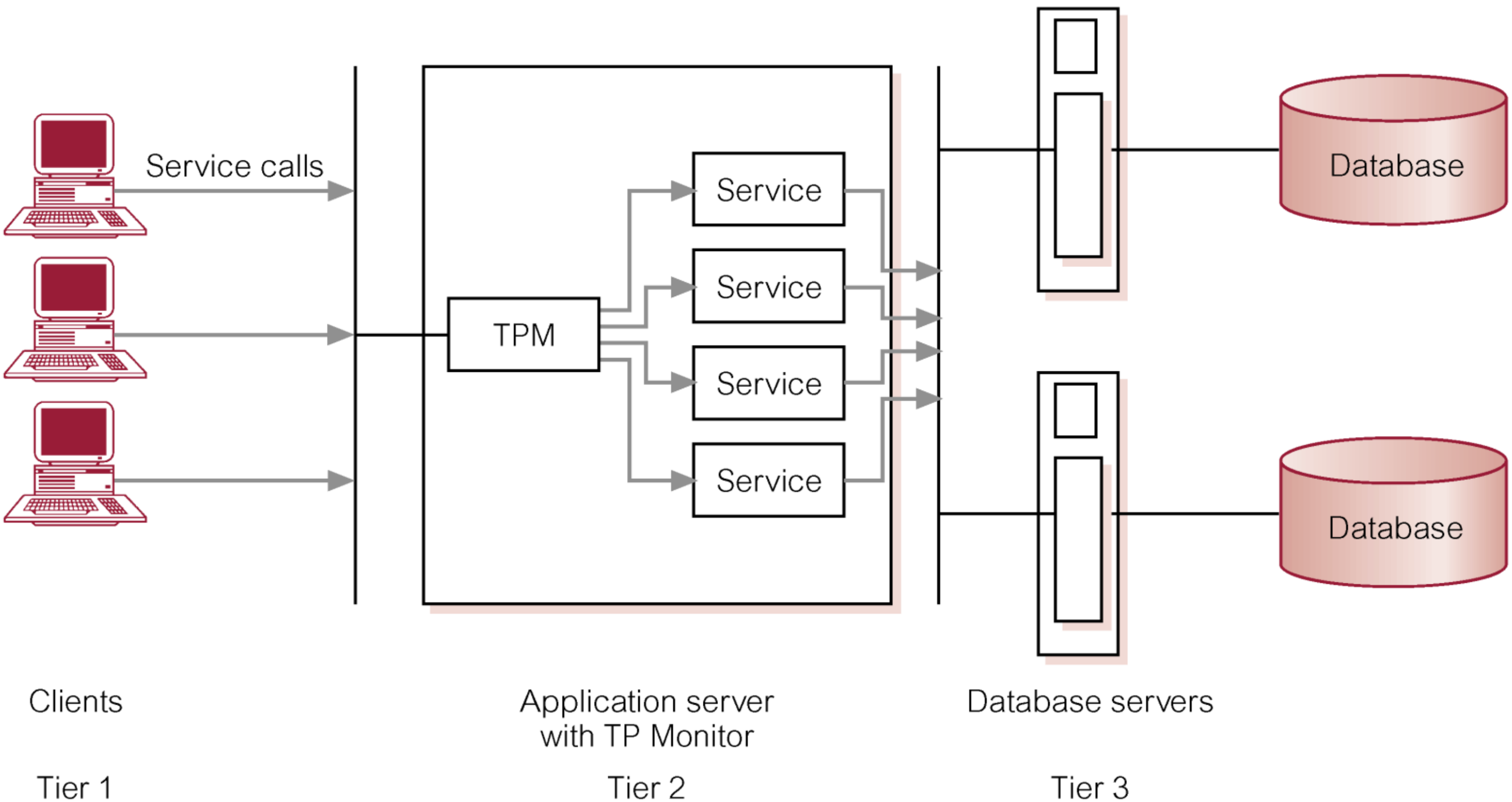
Three-Tier Client-server Architectures



1.6 Transaction Processing Monitors

- Program that controls data transfer between clients and servers in order to provide a consistent environment, particularly for Online Transaction Processing (OLTP).

Transaction Processing Monitor as middle tier of a three-tier client-server architecture



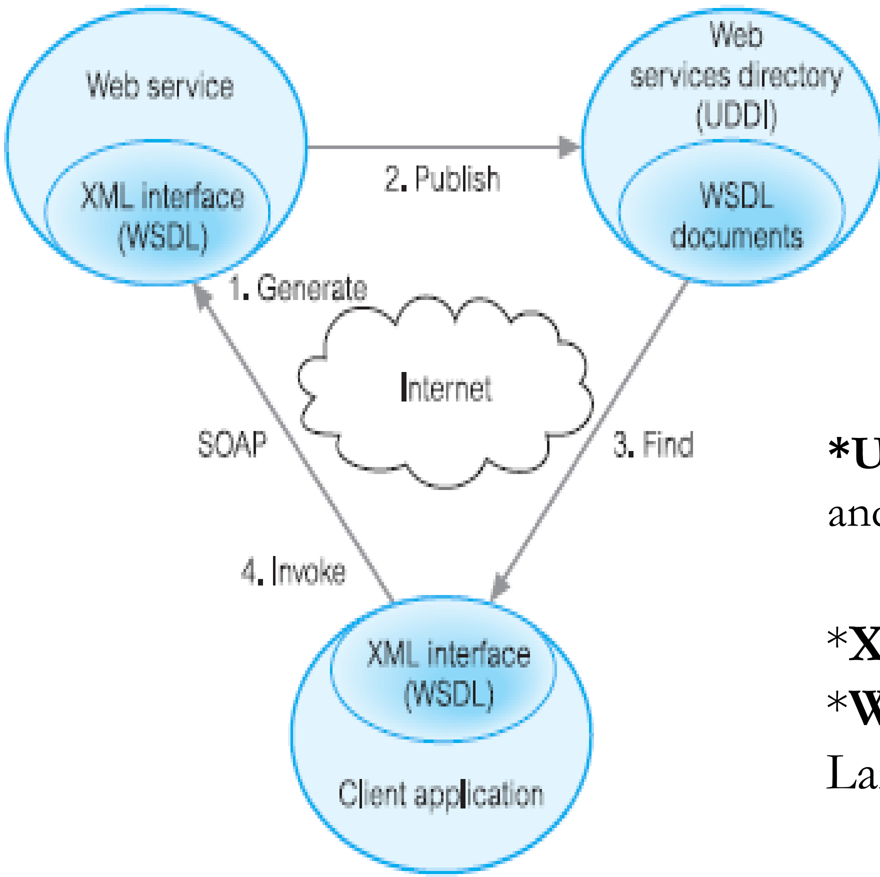
2. Web services and Service-oriented Architectures

- **Service-oriented Architectures**

  Software system that supports interoperable machine-to-machine interaction over a network

  - No user interface
  - For example: Web services
  - Uses widely accepted technologies and standards

Relationship between WSDL,UDDI, and SOAP



* **SOAP**: Simple Object Access Protocol

***UDDI**: Universal Discovery, Description and Integration

***XML**: eXtensible Markup Language
***WSDL**: Web Services Description Language

3. Distributed DBMS

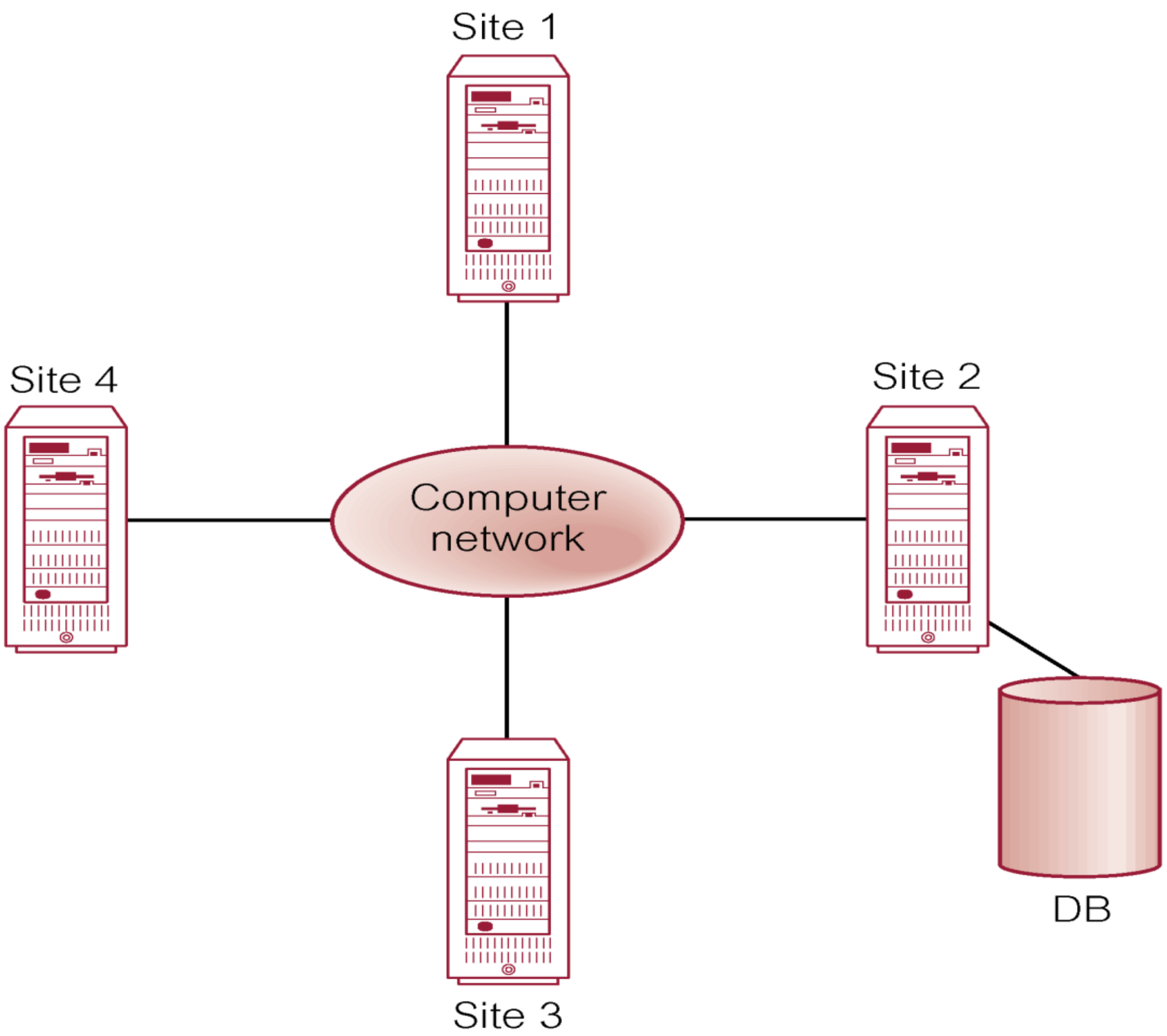
- Distributed database

  Logically interrelated collection of shared data, physically distributed over a computer network.

- Distributed DBMS

  Software system that permits the management of the distributed database. And makes the distribution transparent to users.

Distributed DBMS



Characteristics of Distributed DBMS

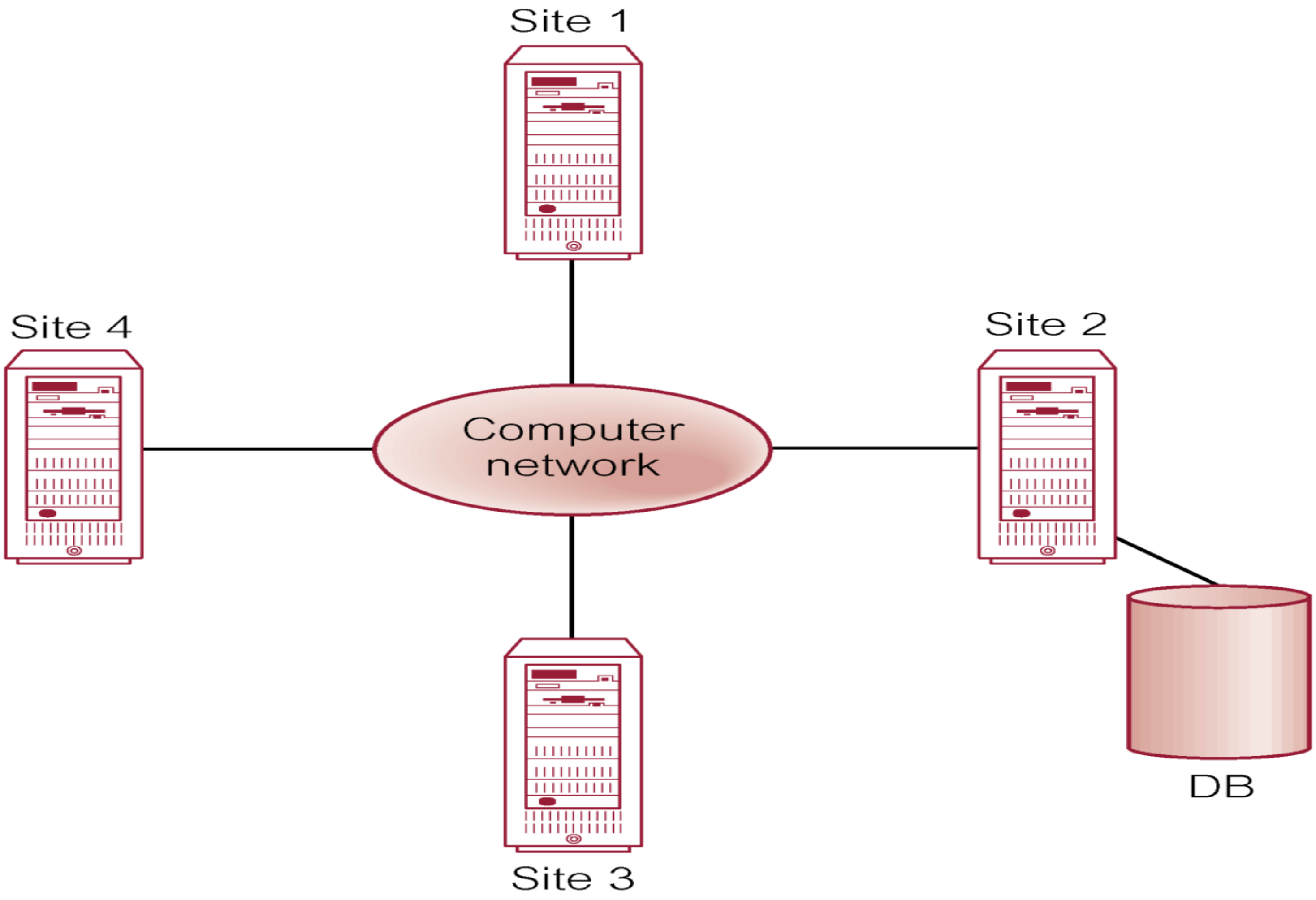
- Collection of logically related shared data
- Data split into fragments
- Fragments may be replicated
- Fragments/replicas are allocated to sites
- Sites are linked by a communications network
- Data at each site is controlled by DBMS
- DBMS handles local apps autonomously
- Each DBMS in one or more global app

Distributed Processing

The difference between Distributed DBMS and Distributed Processing:

- In Distributed DBMS, system consists of data is physically distributed across a number of sites in the network,
- In Distributed Processing, if the data is centralized, even though other users may be accessing the data over the network, will be treated as Distributed Processing and do not consider to be Distributed DBMS.

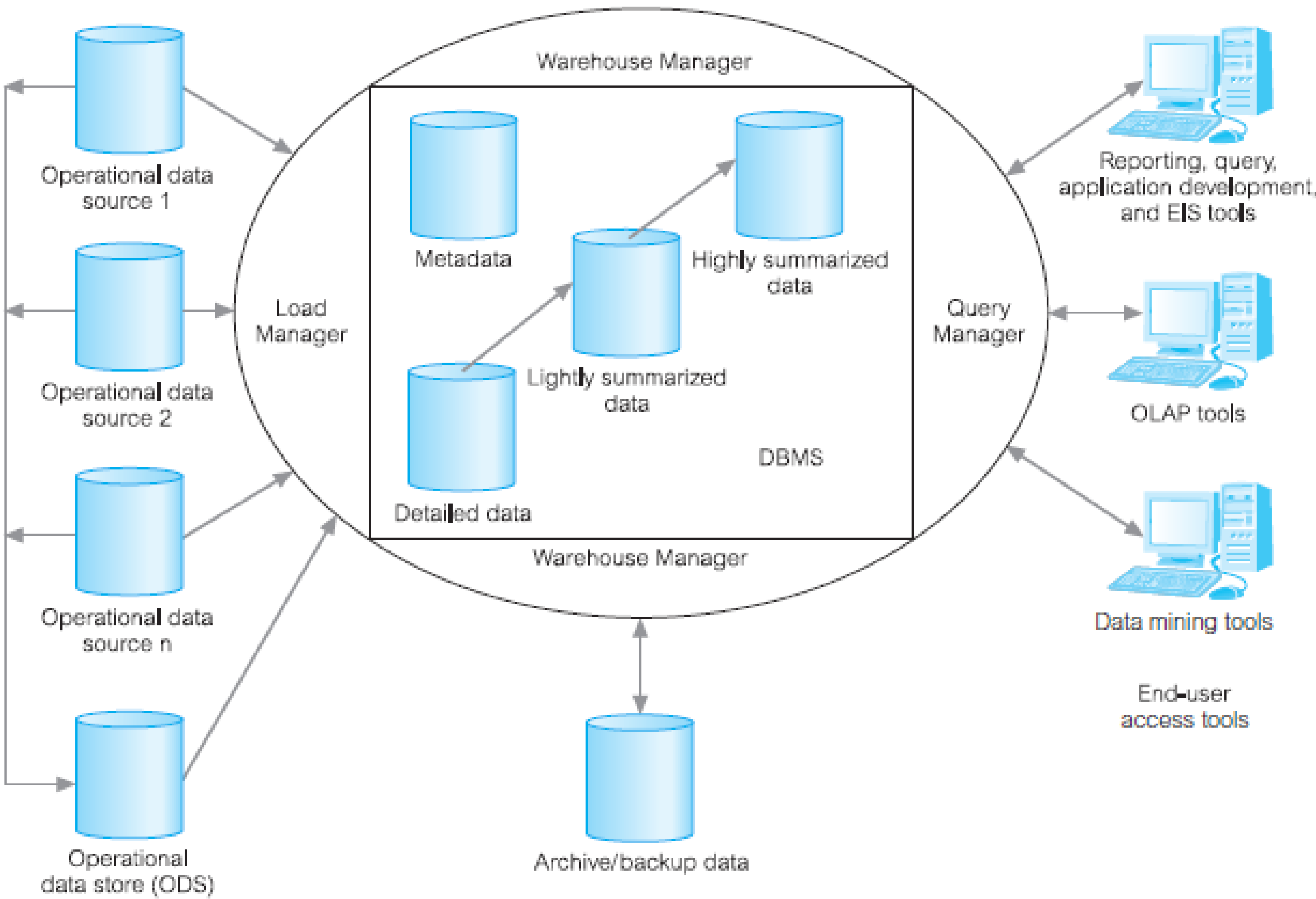
Distributed Processing



4. Data warehousing

- Consolidated/integrated view of corporate data
- Drawn from disparate operational data sources
- Range of end-user access tools capable of supporting simple to highly complex queries to support decision making
- Subject-oriented, integrated, time-variant, and nonvolatile

Typical Architecture of a Data Warehouse

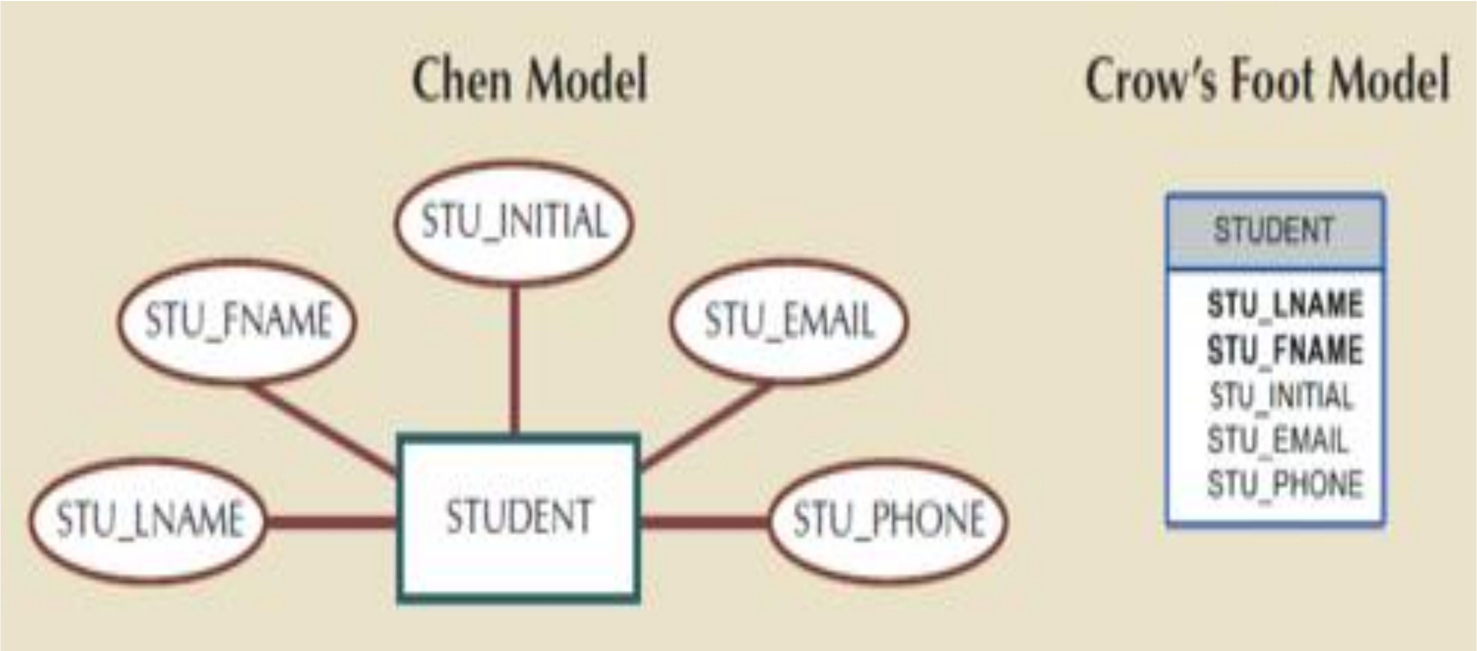


DATA MODELLING USING THE ENTITY-RELATIONSHIP(ER) MODEL

Entity-Relationship (ER) Model

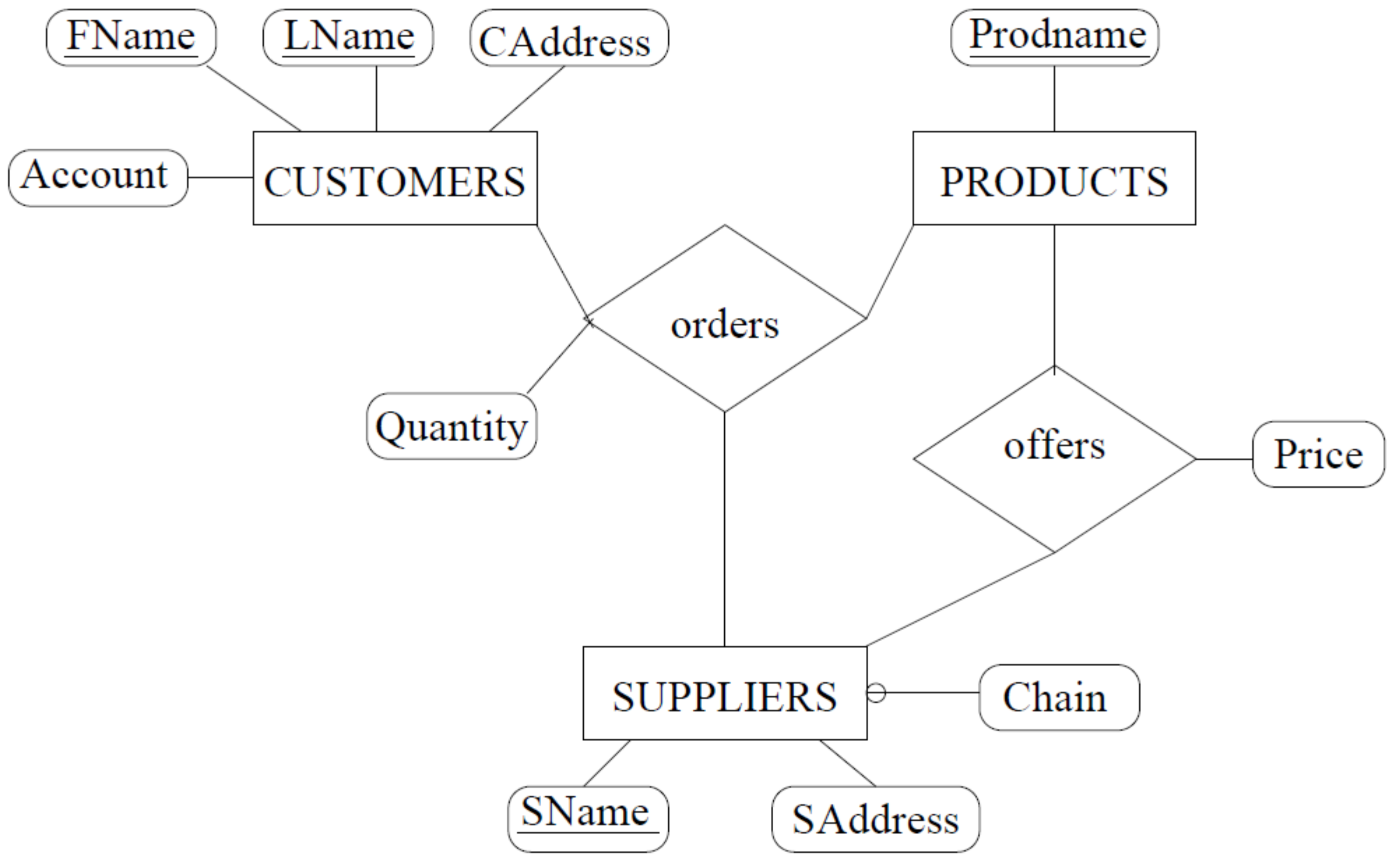
- It is a basis of an Entity Relationship Diagram (ERD).
- Top-down approach to database design that begins by identifying the important data called:
  - *Entities*
  - *Relationships*
  - *Attributes*
  - *Constraints*
- ER concepts can be represented pictorially in an ER Diagram using Unified Modelling Language (UML).

Entity-Relationship (ER) Model



The Attributes of The Student Entity: Chen and Crow's Foot
Source: Coronel & Morris

Entity-Relationship (ER) Model

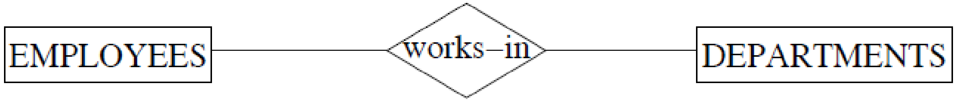


Entity-Relationship (ER) Model

- **Rectangles** represent entity types
- **Ellipses** represent attributes
- **Diamonds** represent relationship types
- **Lines** link attributes to entity types and entity types to relationship types
- **Primary key** attributes are underlined

Entity-Relationship (ER) Model

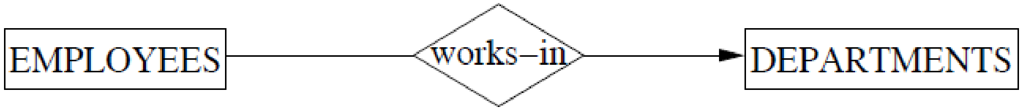
- *Many-To-Many* (default)



Meaning: An employee can work in many departments (≥ 0), and a department can have several employees

Entity-Relationship (ER) Model

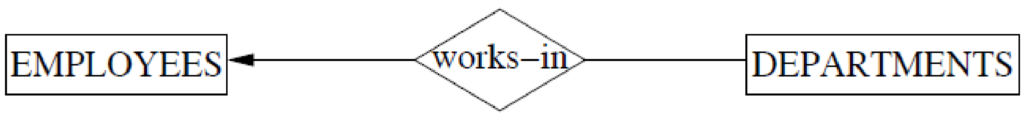
- *Many-To-One*



Meaning: An employee can work in at most one department (≤ 1), and a department can have several employees.

Entity-Relationship (ER) Model

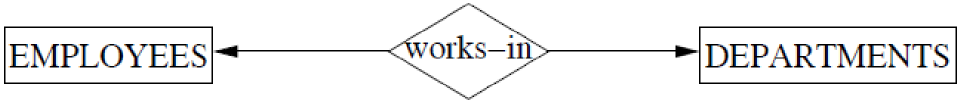
- *One-To-Many*



Meaning: An employee can work in many departments (≥ 0),
but a department can have at most one employee.

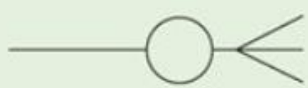
Entity-Relationship (ER) Model

- *One-To-One*



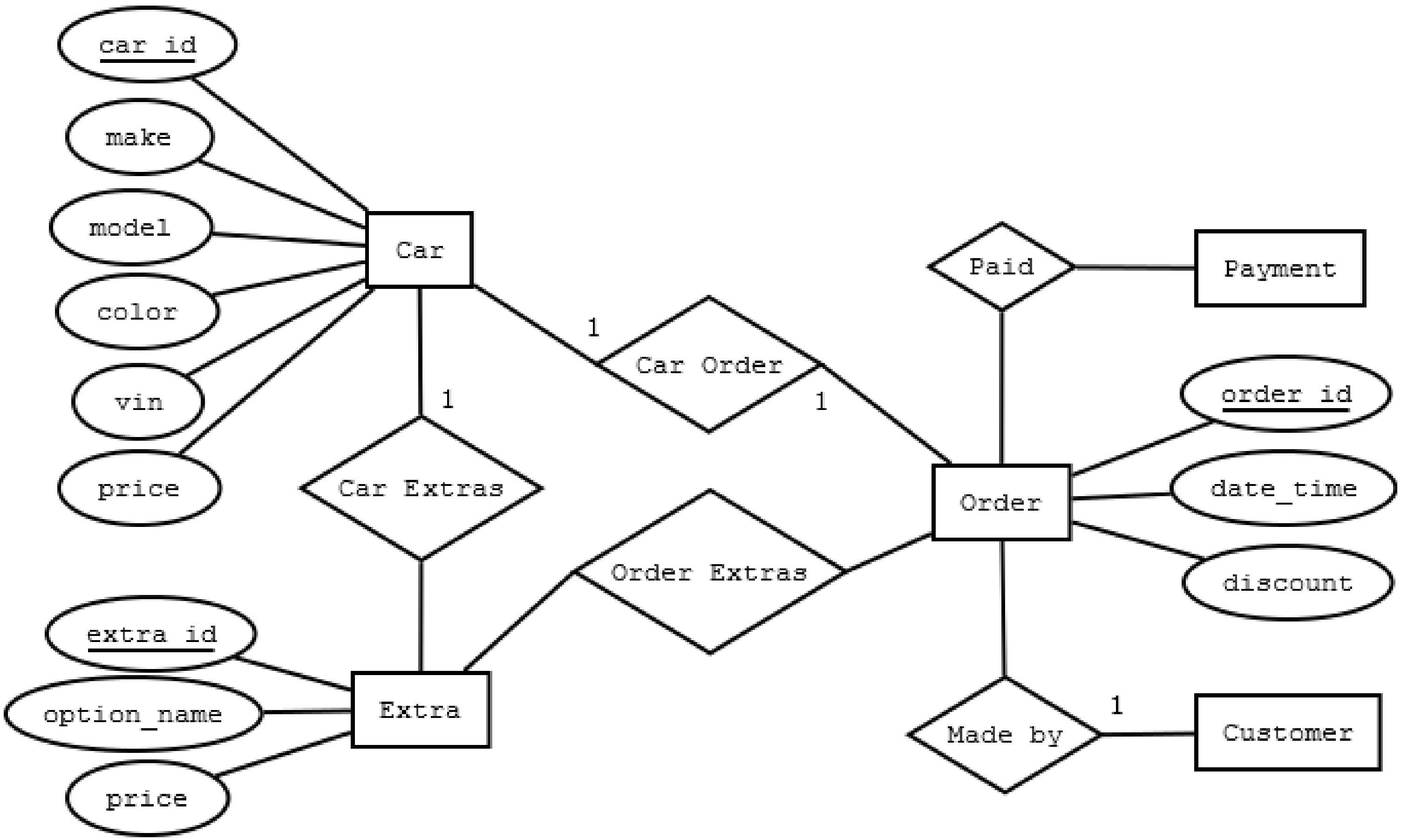
Meaning: An employee can work in at most one department,
and a department can have at most one employee.

Entity-Relationship (ER) Model

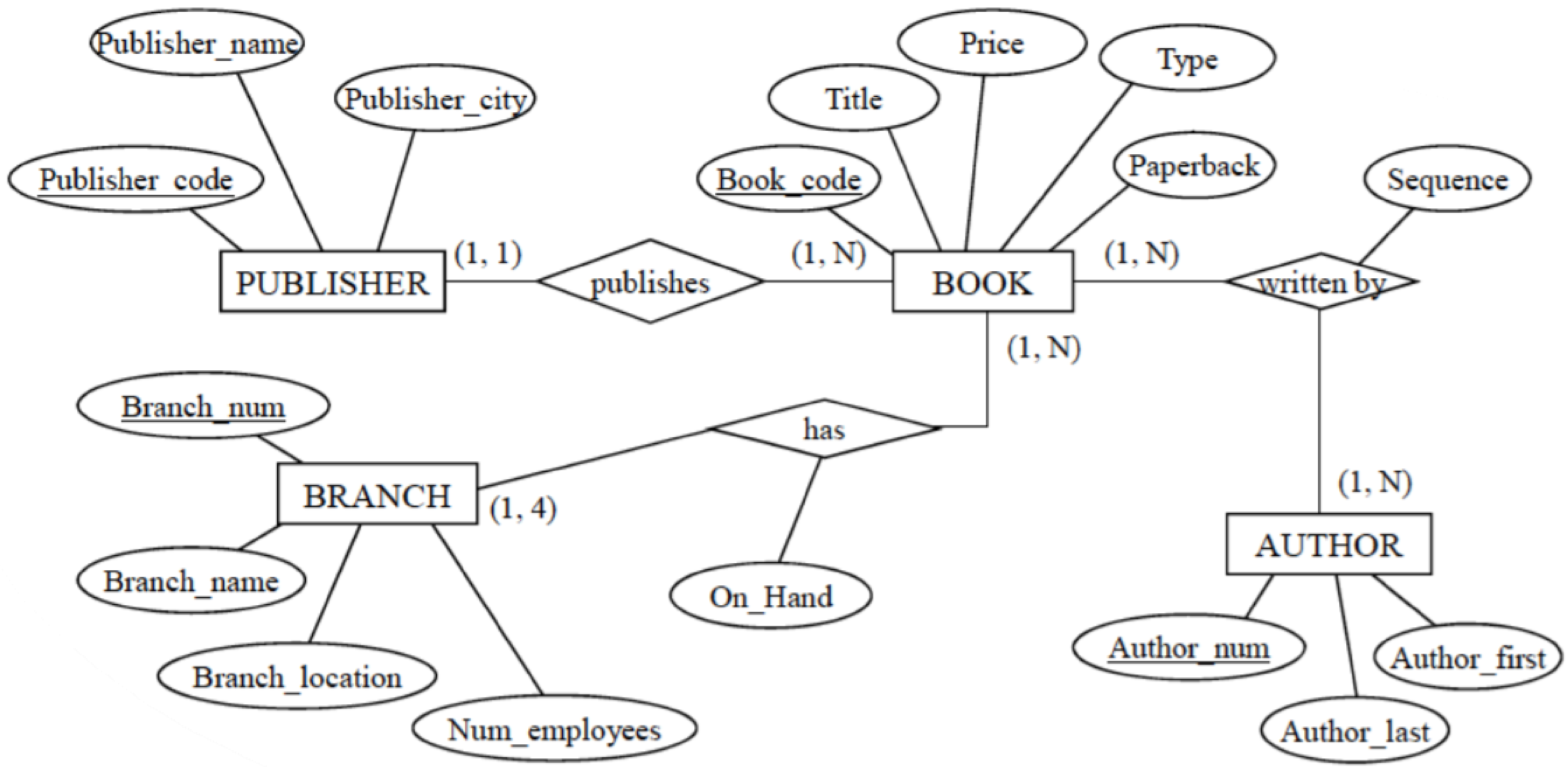
Symbol	Meaning	Numeric Meaning
	Mandatory—One	Exactly one
	Mandatory—Many	One or more
	Optional—One	Zero or one
	Optional—Many	Zero or more

Crow's Foot Notation

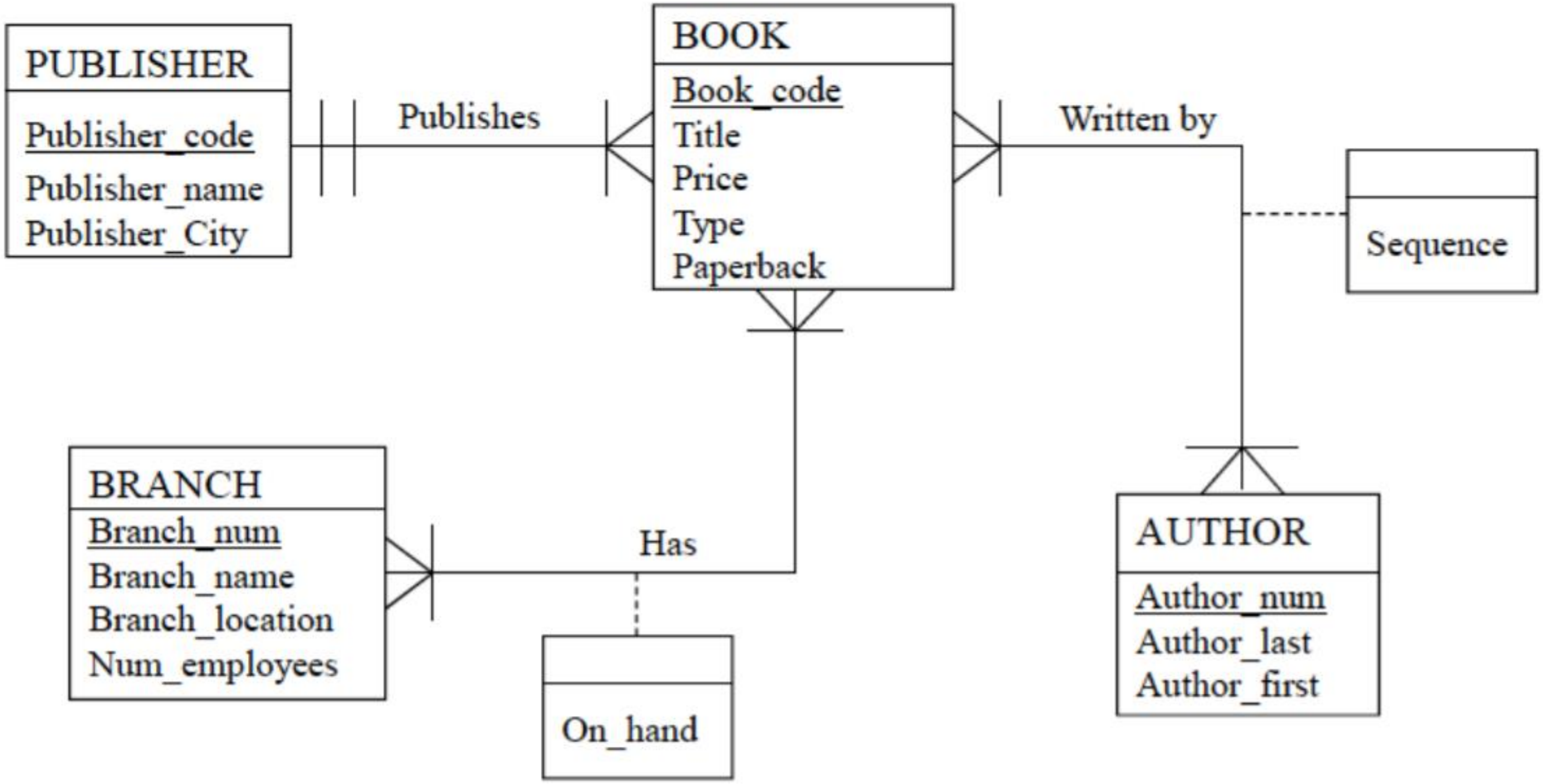
Entity-Relationship (ER) Model



Entity-Relationship (ER) Model



Entity-Relationship (ER) Model



Note: Please use Crow's Foot Model in your Quiz, Test, Assignment and Examination.

The End of Lecture 1
Thank You